

SAMITHA BANDARA

Final-Year Undergraduate, Electronic and Telecommunication Engineering, University of Moratuwa

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RESEARCH INTERESTS

Deep learning, reinforcement learning, robotics, and computer vision.

EDUCATION

University of Moratuwa, Sri Lanka

Expected May 2027

B.Sc. Engineering Honours in Electronic and Telecommunication Engineering

CGPA: **3.82 / 4.00** · Dean's List: Semesters 1, 2, 3, and 6

Nalanda College, Colombo

2022

GCE Advanced Level, Physical Science

3 A grades · Z-score 2.3862 · Island rank 278

RESEARCH EXPERIENCE

Research Assistant, Singapore Management University

Jan 2026 – Aug 2026

School of Computing and Information Systems (SCIS), supported by the SMART M3S programme

- Supervision: Prof. Archan Misra, Dr. Dulanga Weerakoon, and Prasenjit Karmakar (PhD candidate, IIT Kharagpur).
- Researched KV-cache compression and reuse in vision-language models for multi-turn visual question answering.
- Evaluated multimodal models on visual question-answering benchmarks.
- Two manuscripts under review: first author on one, co-author on the other.

RESEARCH PROJECTS

Signal-Processing Techniques for KV-Cache Compression and Reuse in Vision-Language Models for Multi-Turn Visual Question Answering

SMU SCIS, SMART M3S, and NUS. Two manuscripts under review.

Equivariant Attention Reuse for 4D Point-Cloud VLMs under Camera Motion

Ongoing

Reinforcement Learning-Based Dynamic Coupling Compensation and Vision-Guided Grasping for a Hexacopter Aerial Manipulator

Ongoing final-year project. Supervised by Prof. Rohan Munasinghe (University of Moratuwa), in collaboration with Prof. Rajat Talak (National University of Singapore). Developed in NVIDIA Isaac Sim and Isaac Lab, for a real hexacopter with a robotic arm.

SELECTED PROJECTS

Transformer Optimization

GitHub

Adapters, quantization, KV caching, and efficiency techniques for transformer models.

CoreLlama

GitHub

A Llama 2 implementation from scratch in PyTorch.

VLM-gamma

GitHub

A vision-language model built from scratch with a Vision Transformer and transformer decoder.

gpt2-lite

GitHub

A GPT-2 reproduction with a custom training pipeline and BPE tokenization.

TECHNICAL SKILLS

Programming

Python, C, C++, MATLAB

Machine learning

PyTorch, Hugging Face, NumPy, OpenCV

Robotics

Reinforcement learning, Isaac Sim, Isaac Lab, Gazebo, PPO

Systems

Linux, Slurm, Docker, Git

Embedded systems

STM32, embedded C, PCB design, sensors